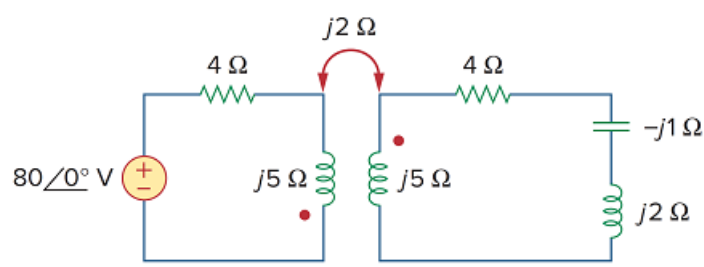


Problem

For the circuit in Fig. 13.82, determine the impedance seen by the source.

Figure 13.82



Step-by-step solution

Step 1 of 3

Refer to Figure 13.82 in the textbook.

Redraw Figure 13.82 in the textbook, with loop currents.

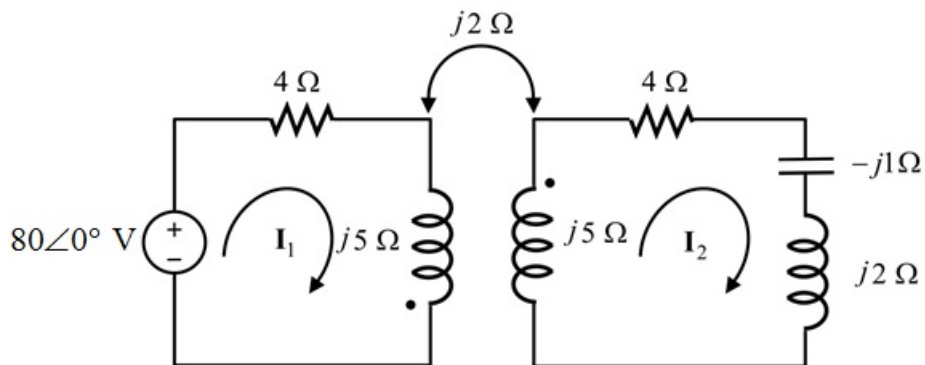


Figure 2

Step 2 of 3

Calculate the value of the impedance $\mathbf{Z}_s$ seen by the source.

Write the expression for the impedance $\mathbf{Z}_s$.

$$\mathbf{Z}_s = \frac{\mathbf{V}_s}{\mathbf{I}_1}$$
$$\mathbf{Z}_s = \frac{80\angle 0^\circ}{\mathbf{I}_1}$$

Apply Kirchhoff's Voltage Law in mesh-1.

$$(4 + j5)\mathbf{I}_1 + j2\mathbf{I}_2 = 80\angle 0^\circ$$

$$(4 + j5) + j2\frac{\mathbf{I}_2}{\mathbf{I}_1} = \frac{80\angle 0^\circ}{\mathbf{I}_1}$$

$$\frac{80\angle 0^\circ}{\mathbf{I}_1} = (4 + j5) + j2\frac{\mathbf{I}_2}{\mathbf{I}_1}$$

Substitute $\mathbf{Z}_s$ for $\frac{80\angle 0^\circ}{\mathbf{I}_1}$ in this expression.

$$\mathbf{Z}_s = (4 + j5) + j2\frac{\mathbf{I}_2}{\mathbf{I}_1} \dots\dots (1)$$

Step 3 of 3

Apply Kirchhoff's Voltage Law in mesh-2.

$$j2\mathbf{I}_1 + (j5 + 4 - j1 + j2)\mathbf{I}_2 = 0$$

$$j2\mathbf{I}_1 + (4 + j6)\mathbf{I}_2 = 0$$

$$\frac{\mathbf{I}_2}{\mathbf{I}_1} = \frac{-j2}{(4 + j6)} \dots\dots (2)$$

Substitute equation (2) in equation (1).

$$\mathbf{Z}_s = (4 + j5) + j2\left(\frac{-j2}{(4 + j6)}\right)$$
$$= (4 + j5) + \frac{4}{(4 + j6)}$$
$$= 4 + j5 + 0.3076 - j0.4615$$
$$= 4.308 + j4.538 \, \Omega$$

Therefore, the value of impedance $\mathbf{Z}_s$ seen by the source is $\boxed{4.308 + j4.538 \, \Omega}$.